

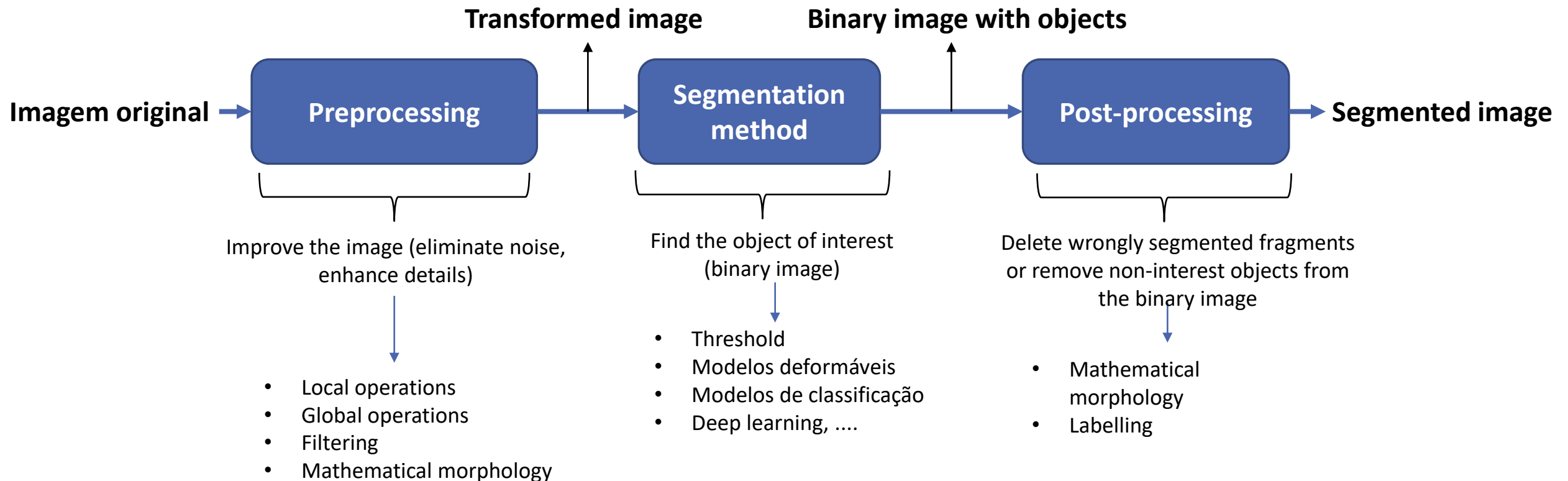
Biomedical Image Processing

MATHEMATICAL MORPHOLOGY

Helena R. Torres

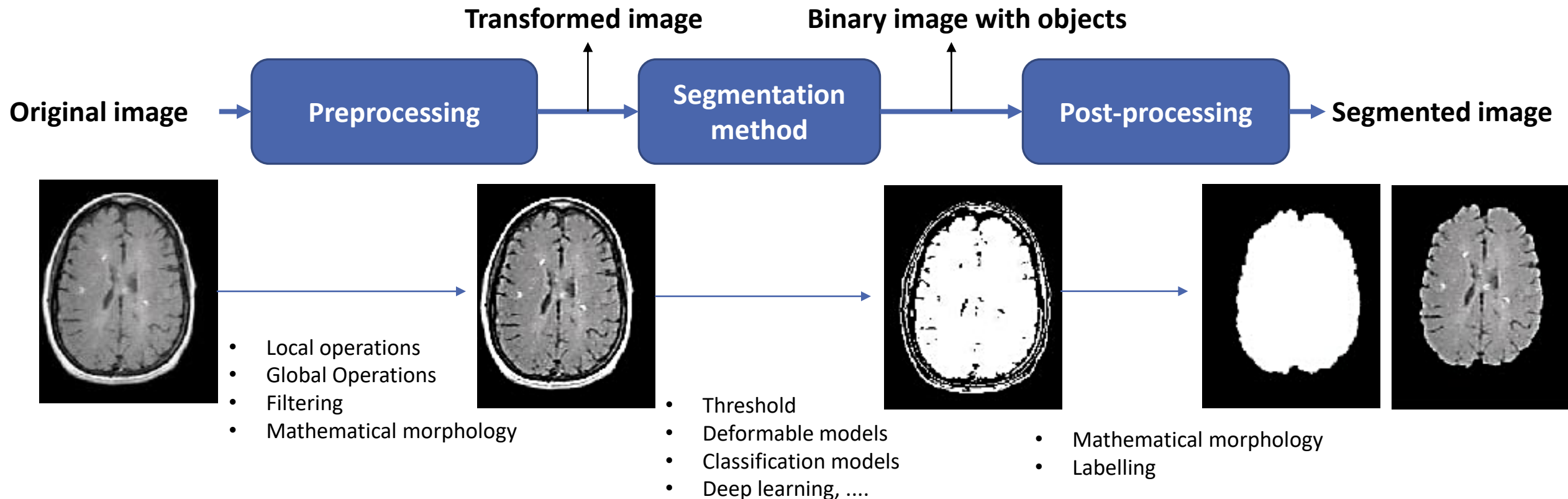
Medical Image Processing

PIPELINES FOR DELINEATION OR DETECTION OF ANATOMICAL STRUCTURES IN MEDICAL IMAGING



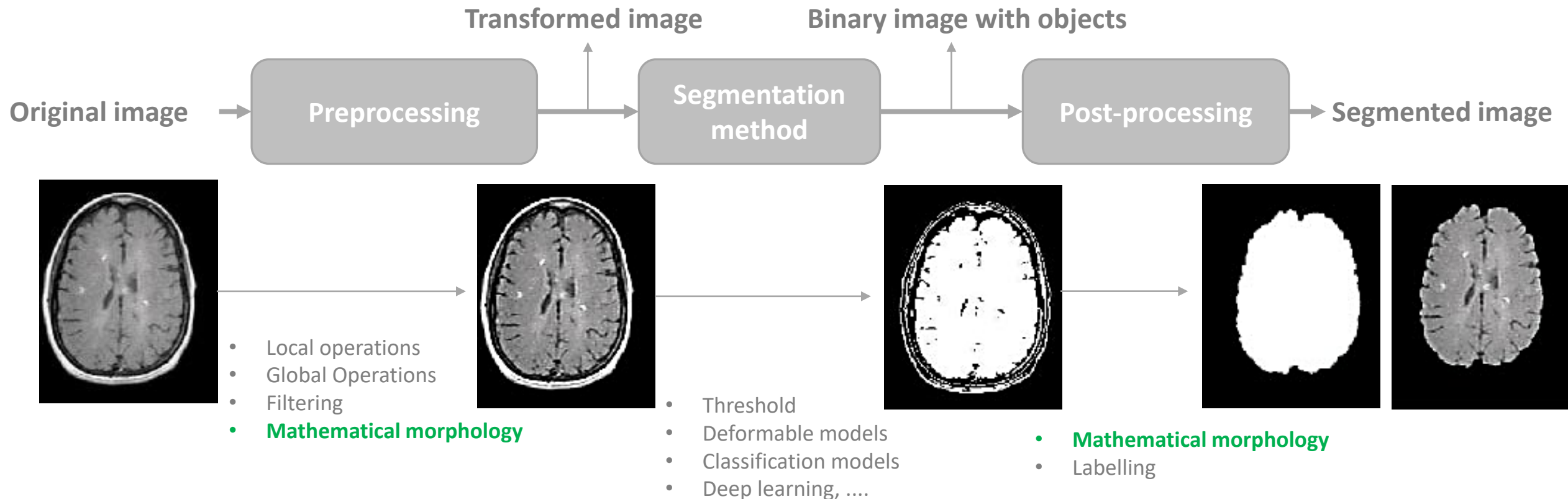
Medical Image Processing

PIPELINES FOR DELINEATION OR DETECTION OF ANATOMICAL STRUCTURES IN MEDICAL IMAGING



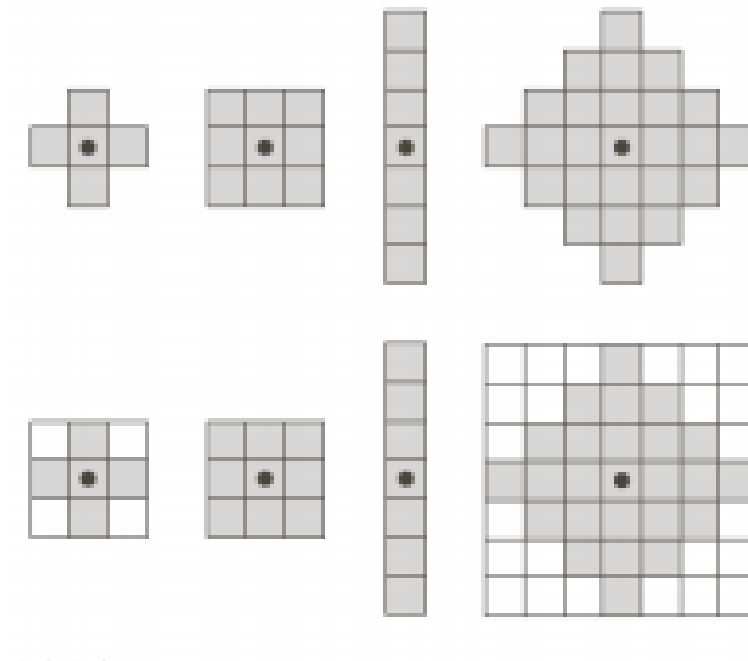
Medical Image Processing

PIPELINES FOR DELINEATION OR DETECTION OF ANATOMICAL STRUCTURES IN MEDICAL IMAGING



Mathematical morphology

- ❑ An instrument that acts on the shape and structure properties of areas of an image for extracting objects from the image
- ❑ Nonlinear operations to act on objects
- ❑ It requires:
 - the image to process
 - a structuring element (*kernel*)
 - 3x3, 7x7, ...

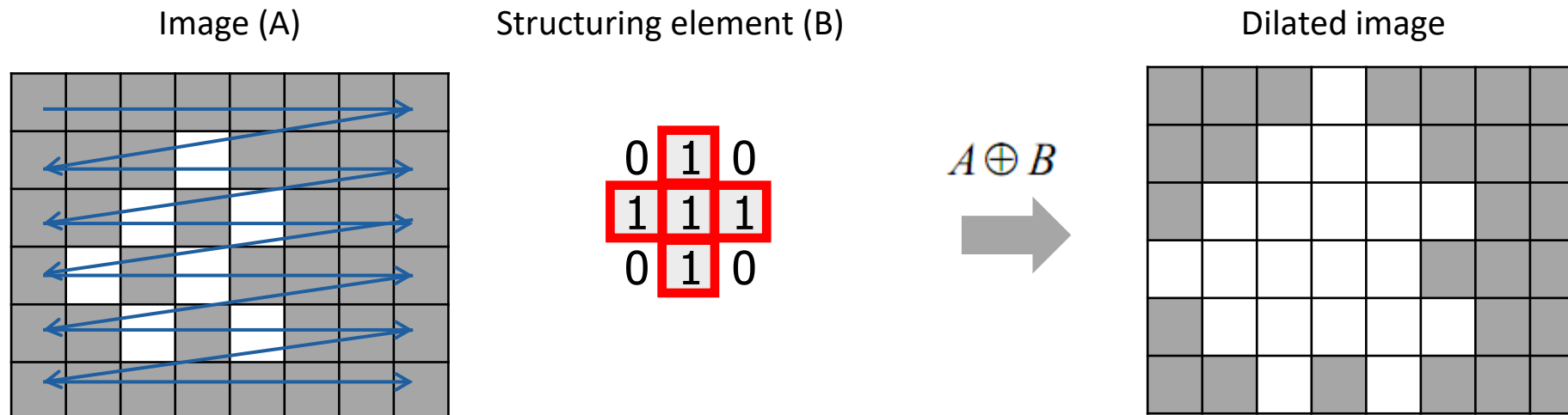


Mathematical morphology

BINARY DILATION

$$A \oplus B$$

- Add pixels to the boundaries of a segmented region, increasing its area and filling in some zones inside
- Center B on each background pixel. If there is at least one foreground pixel in the region defined by B, then the pixel is considered foreground.

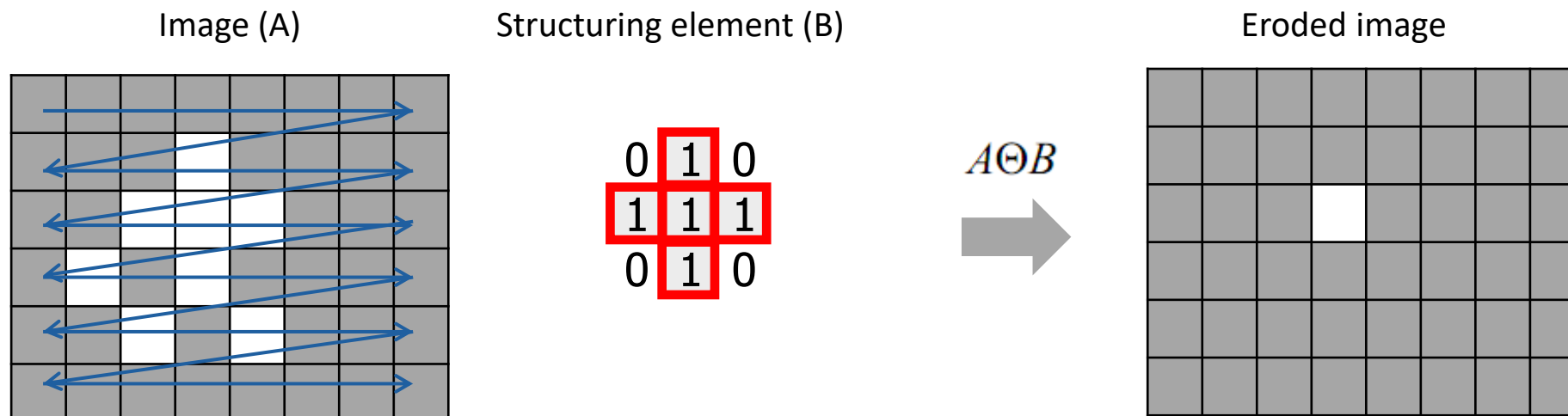


Mathematical morphology

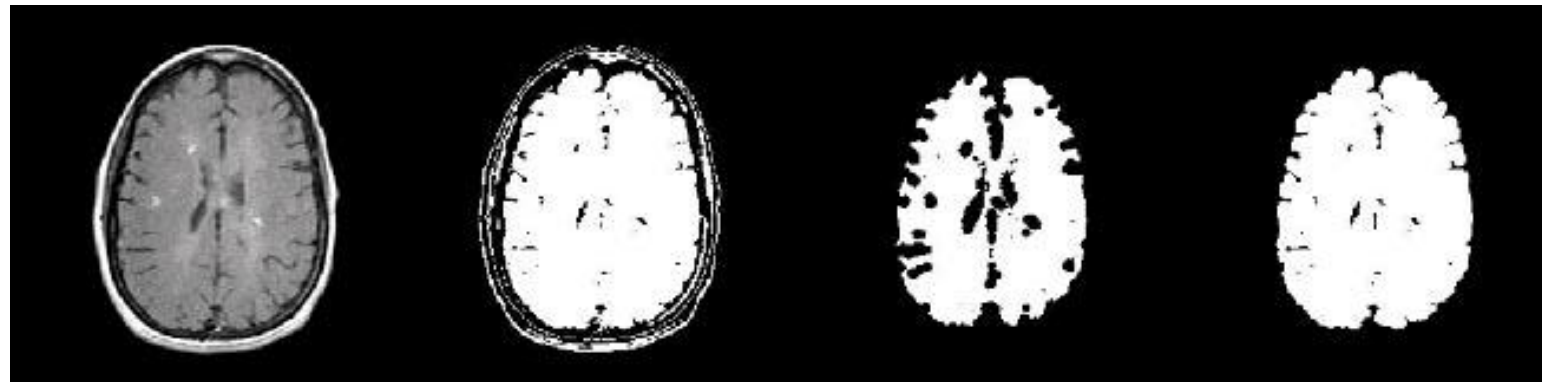
BINARY EROSION

$$A \ominus B$$

- ❑ Remove pixels to the boundaries of a segmented region, decreasing its area and eliminating regions of size inferior to the structuring element
 - Center B on each foreground pixel. If there is at least one background pixel in the region defined by B, then the pixel is eliminated.



Mathematical morphology



Original image

Binarized image

Eroded image

Dilated image

Mathematical morphology

BINARY OPENING AND CLOSING

❑ Combination of erosion and dilation operators

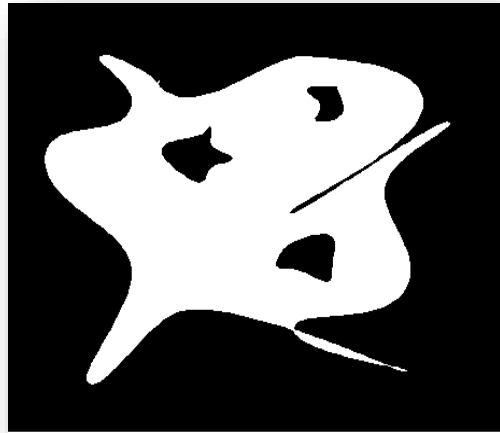
❑ Opening

- Erosion followed by a dilation $A \circ B = (A \ominus B) \oplus B$
- Remove small regions

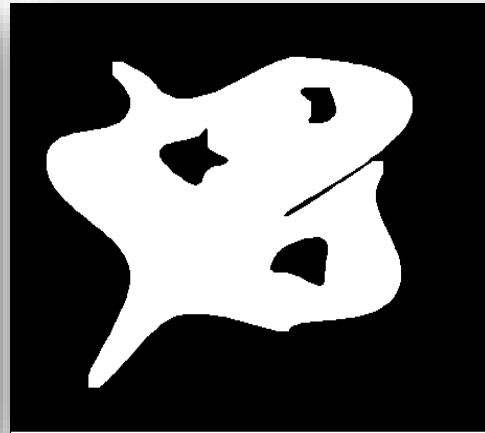
❑ Closing

- Dilation followed by erosion $A \bullet B = (A \oplus B) \ominus B$
- Gap fill

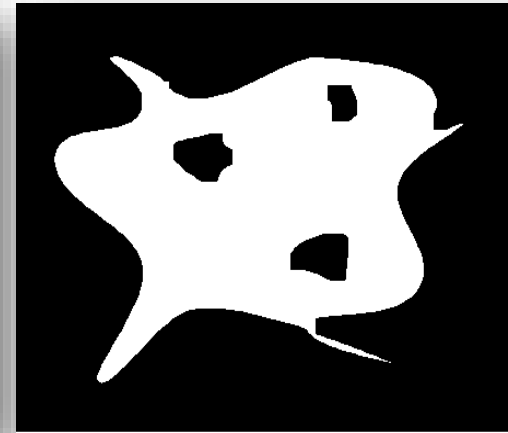
Mathematical morphology



Original image



Opening



Closing

Mathematical morphology

DILATION AND EROSION - GRAYSCALE

□ Dilation

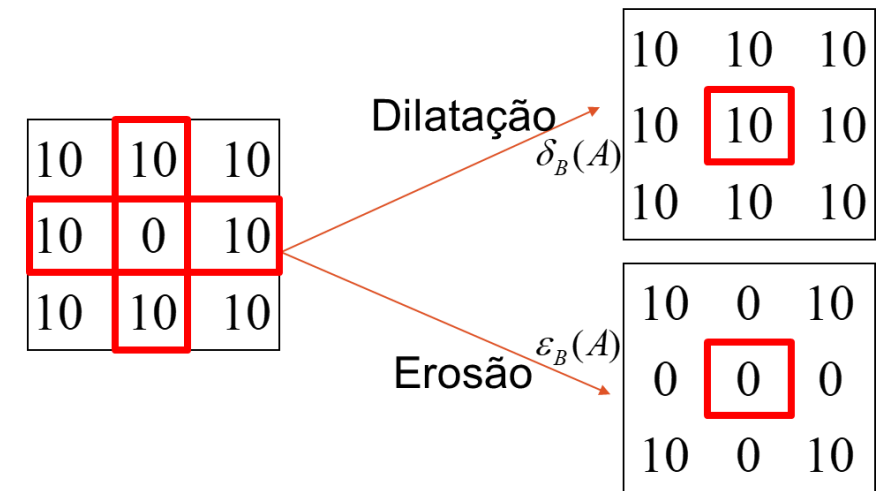
- Calculates the maximum value of image A, in the region defined by the structuring element B, and assigns this value to the central pixel of the region.

$$\delta_B(A) = \max\{x_k, k \in B\}$$

□ Erosion

- Calculates the minimum value of image A, in the region defined by the structuring element B, and assigns this value to the central pixel of the region.

$$\varepsilon_B(A) = \min\{x_k, k \in B\}$$



Mathematical morphology



Erosion



Dilation



Opening



Closing



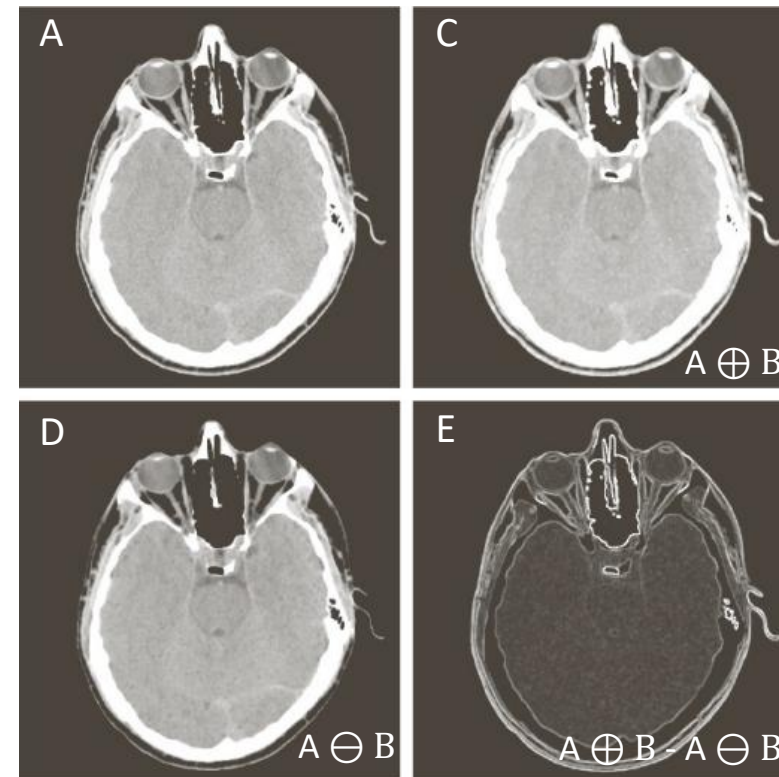
Mathematical morphology

❑ Morphological operations can be used to construct filters similar to spatial filters:

❑ Contour Detection and Morphological Gradient



A: original image; C: Outline (A-Eroded);

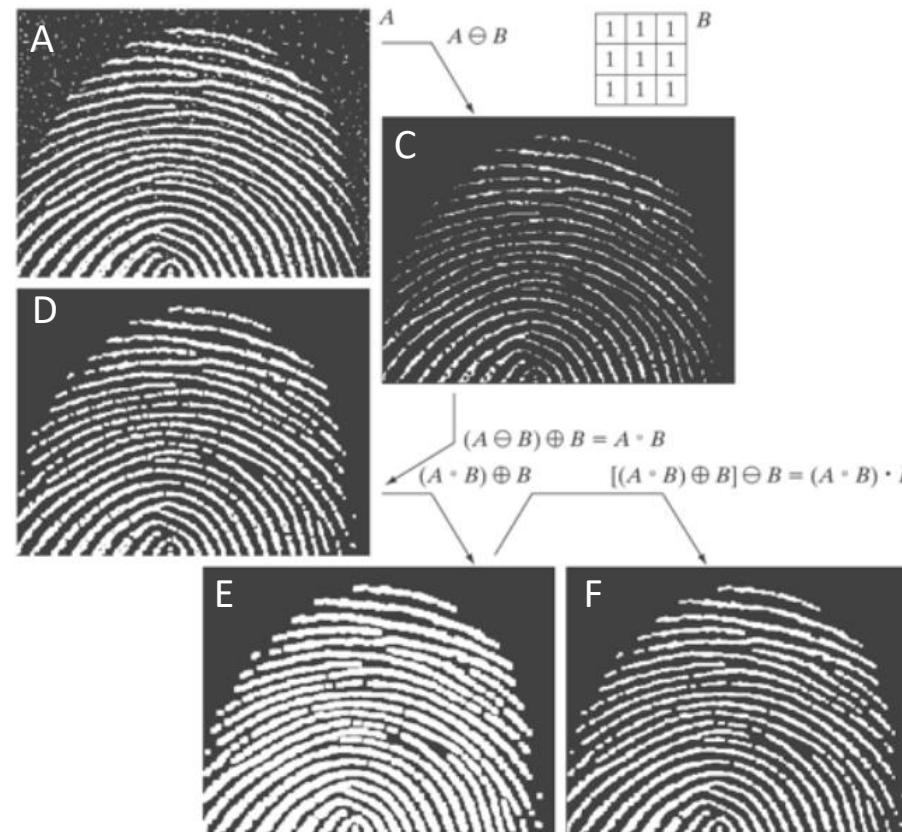


A: original image;
B: Structuring element;
C: Dilation;
D: Erosion;
E: Morphological gradient (C-D)

Mathematical morphology

❑ Morphological operations can be used to construct filter similar to spatial filters:

❑ Noise removal

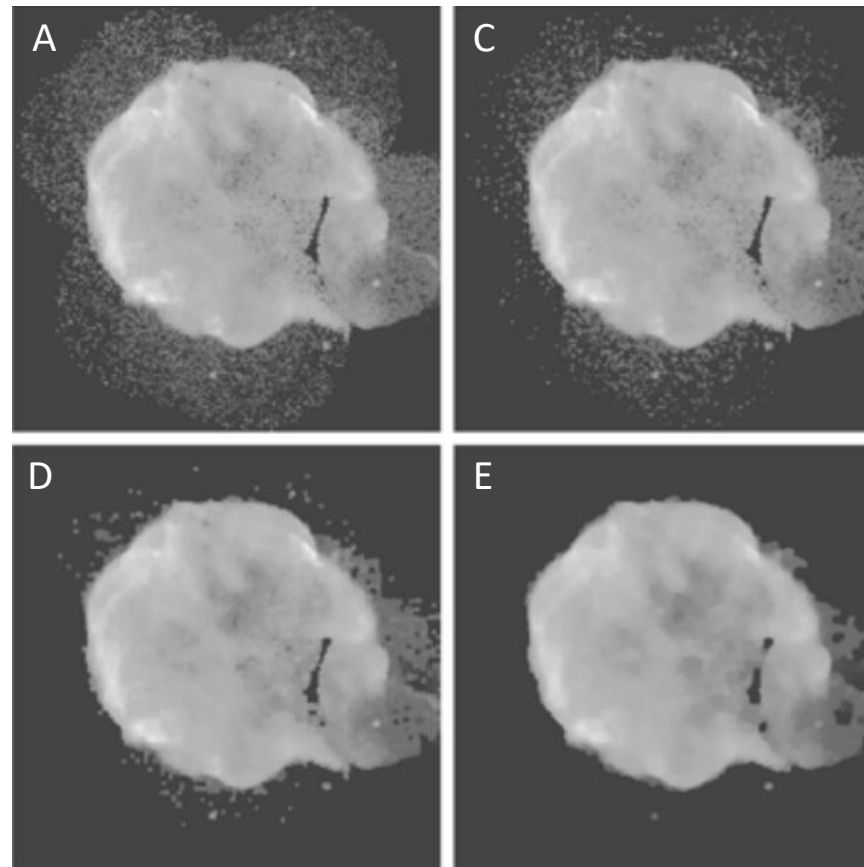


A: original image;
B: Structuring element;
C: Erosion;
D: Opening of A;
E: Dilation of D
F: Closing of E

Mathematical morphology

- ❑ Morphological operations can be used to construct filter similar to spatial filters:

- ❑ Morphological smoothing



A: original image;

C: Opening and closing with small structuring element

D: Opening and closing with medium structuring element

E: Opening and closing with large structuring element

Mathematical morphology



APPLICATIONS

☐ Post-processing of segmented images

- ☐ Hole filling
- ☐ Convex hull
- ☐ Thinning/Thickening
- ☐ Skeletonization
- ☐ Noise removal, hole closure

☐ Edge detection

☐ Morphological smoothing

☐ Morphological gradient

Binary morphology

Morphology at gray levels

`skimage.morphology`

Morphological algorithms, e.g., closing, opening, skeletonization.

<code>area_closing</code>	Perform an area closing of the image.
<code>area_opening</code>	Perform an area opening of the image.
<code>ball</code>	Generates a ball-shaped footprint.
<code>binary_closing</code>	Return fast binary morphological closing of an image.
<code>binary_dilation</code>	Return fast binary morphological dilation of an image.
<code>binary_erosion</code>	Return fast binary morphological erosion of an image.
<code>binary_opening</code>	Return fast binary morphological opening of an image.
<code>black_tophat</code>	Return black top hat of an image.
<code>closing</code>	Return grayscale morphological closing of an image.

Mathematical morphology

EXAMPLE

- Framework to obtain a model of a bone through the application of morphological operations



Original image (img)



Image after threshold (imgTh)

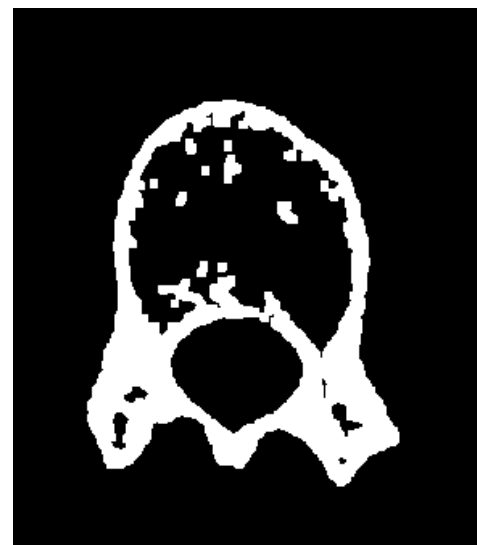


Image after opening (imgA)



Image after hole filling (imgFinal)

Postprocessing

POSTPROCESSING

- ❑ Mathematical morphology
- ❑ Choosing the relevant blobs – Labeling

LABELLING

- ❑ After the image is segmented, the binary image can contain multiple objects.
- ❑ Thus, it is necessary to tag each pixel, according to the region to which it belongs (blob).
- ❑ After each blob has been tagged, you can analyze its characteristics to find out which object of interest:
 - ❑ Size, bounding box properties, circularity, ...



Postprocessing

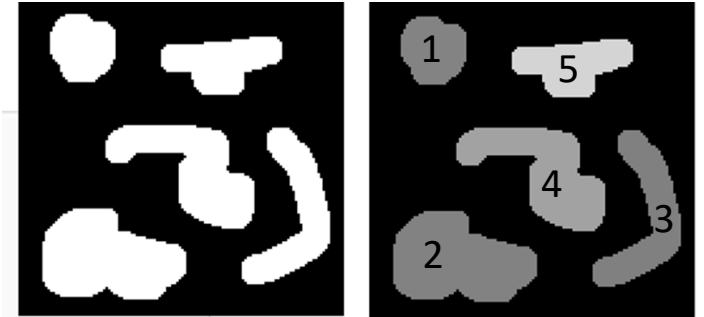
LABELLING

- ❑ Function for tagging blobs in a binary image:

- ❑ `from skimage.measure import label`
- ❑ This function assigns a class to each object

- ❑ Function that calculates various characteristics of connected objects (blobs):

- ❑ `from skimage.measure import regionprops`
- ❑ It has as input an image with classes (resulting from the previous step) and generates the characteristics of each object
- ❑ Important measures:
 - ❑ area – number of pixels of the object;
 - ❑ bbox – position and size of the smallest box that includes the object;
 - ❑ centroid – center of mass of the object;
 - ❑ eccentricity – How close the object is to a circle;
 - ❑ Perfect circle – 0
 - ❑ orientation.



```
from skimage.measure import label, regionprops
label_img = label(img)
regions = regionprops(label_img)

print(regions[0].area) #print area of the first object

for props in regions: #loop to print centroid of objects
    print(props.centroid[0])
    print(props.centroid[1])
```

Postprocessing

EXAMPLE – POST-PROCESS A SEGMENTATION RESULT

